
Publishing FAIR and Machine-actionable Reviews in Materials Science: The Case for Symbolic Knowledge in Neuro-symbolic Artificial Intelligence

Jennifer D’Souza

TIB Leibniz Information Centre for Science and Technology
Hannover, Germany
jennifer.dsouza@tib.eu

Sören Auer

TIB Leibniz Information Centre for Science and Technology
Hannover, Germany
L3S Research Center, Leibniz University Hannover
Hannover, Germany

Eleni Poupaki

Eindhoven University of Technology (TU/e)
Eindhoven, The Netherlands

Alex Watkins

University of Warwick
Coventry, United Kingdom

Anjana Devi

Leibniz Institute for Solid State and Materials Research (IFW)
Dresden, Germany

Riikka L. Puurunen

School of Chemical Engineering
Aalto University
Espoo, Finland

Bora Karasulu

University of Warwick
Coventry, United Kingdom

Adrie Mackus

Eindhoven University of Technology (TU/e)
Eindhoven, The Netherlands

Erwin Kessels

Eindhoven University of Technology (TU/e)
Eindhoven, The Netherlands

Abstract

Scientific reviews are central to knowledge integration in materials science, yet their key insights remain locked in narrative text and static PDF tables, limiting reuse by humans and machines alike. This article presents a case study in atomic layer deposition and etching (ALD/E) where we publish review tables as FAIR, machine-actionable comparisons in the Open Research Knowledge Graph (ORKG), turning them into structured, queryable knowledge. Building on this, we contrast symbolic querying over ORKG with large language model-based querying, and argue that a curated symbolic layer should remain the backbone of reliable neurosymbolic AI in materials science, with LLMs serving as complementary, symbolically grounded interfaces rather than standalone sources of truth.

“Consider a future device ... in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility. It is an enlarged intimate supplement to his memory.”

— Vannevar Bush, July 1945, *As We May Think* [9]

1 Introduction

Vannevar Bush’s 1945 vision of a personal, searchable “memex”[9] resonates strongly with today’s situation in materials science. Atomic layer deposition and etching (ALD/ALE) now generate vast literature of processes, mechanisms, and performance data, much of it communicated through narrative text and multi-page PDF tables. Review articles help by collating this information, yet answering even a simple comparative question (e.g., “which reported ALD processes for material X meet a given temperature window and growth-per-cycle range?”) often still involves substantial manual effort—reading, reconstructing tables in spreadsheets, and filtering entries. As a result, the extracted synthesis is frequently captured in ad hoc formats (e.g., personal notes or local spreadsheets) that are difficult to share, query, and reuse systematically.

In parallel, there is growing interest in making scientific knowledge *machine-actionable*: structured in such a way that it can be searched, filtered, and recombined automatically, in line with the FAIR Guiding Principles (Findable, Accessible, Interoperable, Reusable) for scientific data [62]. The Open Research Knowledge Graph (ORKG)[6, 7] is one such next-generation infrastructure. Instead of publishing only PDFs, authors and curators can represent key findings as structured “comparisons” that look like familiar tables—rows for individual studies, columns for materials, precursors, substrates, process conditions, and performance metrics—but are stored as a knowledge graph with stable identifiers and an explicit schema. Earlier work has shown how such “SmartReviews” can make survey-article tables both human- and machine-readable[46], and related efforts such as MatKG[61] demonstrate that large, literature-derived knowledge graphs are feasible in materials science.

At the same time, large language models (LLMs) have quickly become attractive tools for materials researchers. Recent work in ALD benchmarking, for example, evaluates GPT-4-class models as assistants for design and conceptual questions about ALD processes[65], while MOF-ChemUnity couples a literature-derived knowledge base with LLM-based assistance for metal–organic framework research[53]. These systems are appealing because they can read PDFs, answer questions in natural language, and help explain trends or generate hypotheses. However, LLMs are also stochastic generative models: they sample from a probability distribution over possible next tokens, so that the same question can yield slightly different answers from run to run. When precise numbers and exact table entries matter—as they do for process windows, growth-per-cycle values, or etch selectivities—this lack of determinism, together with known issues such as hallucinated facts and mis-read tables, limits their suitability as standalone tools for quantitative decision-making[69, 18].

In this work we bring these two developments together for the specific case of ALD/ALE review articles. Our starting point is a symbolic layer: we take nine recent review papers (three in ALD, six in ALE) and convert 18 eligible review tables—those that summarise processes and conditions across multiple primary studies—into machine-actionable ORKG comparisons. Each comparison is a structured, human-curated table with typed columns for materials, precursor chemistries, substrates, process conditions, and performance descriptors, and each row is linked back to the cited reference. On top of this symbolic representation, we then study how well current LLMs can answer precise scientific questions about these tables, both when they only see the original PDFs and when they are explicitly *grounded* in the ORKG comparisons. Across these settings, we find that symbolic querying over ORKG comparisons provides the most precise and reproducible answers for review-style questions, while PDF-only LLM querying frequently struggles with table structure and numerical accuracy. Expert feedback confirms that the machine-actionable comparisons are immediately useful for comparative reasoning in ALD/ALE research. Grounding LLMs in these structured tables substantially improves answer reliability, but does not fully replace symbolic querying as a gold standard.

Concretely, we address three research questions:

- RQ1** *How useful do ALD/ALE experts find machine-actionable ORKG comparison tables for answering complex review questions?* We treat the ORKG tables as a new form of review output and ask practising researchers to rate their meaningfulness and usefulness, and to comment on limitations and wish-list queries.
- RQ2** *To what extent can LLMs, operating purely over review-article PDFs, reproduce the precise, table-style answers obtained from symbolic querying?* Here we upload the PDFs to state-of-the-art LLMs, pose detailed questions that mirror SPARQL queries, and measure how closely the model-generated tables match the ORKG-based gold standard.
- RQ3** *How well can LLMs answer the same questions when explicitly grounded in machine-actionable ORKG comparison tables, and how does this compare to both the symbolic SPARQL baseline and purely PDF-based querying?* In this setting, models receive the exact ORKG comparison (exported as a CSV table) together with the natural-language question, allowing us to test whether structured symbolic context closes part of the gap to symbolic querying.

Our contributions are threefold. First, we publish 18 curated ORKG comparison tables from nine ALD/ALE review articles, together with 33 natural-language questions and their SPARQL formulations, released as the *ALD/E Query Dataset*[15]. Second, we report a domain-expert evaluation in which three ALD/ALE researchers judge the questions and SPARQL-derived tables to be scientifically meaningful and practically useful for comparative reasoning. Third, we benchmark three question-answering regimes over the same content—(i) symbolic SPARQL over ORKG, (ii) PDF-only LLM querying, and (iii) LLM querying grounded in ORKG tables—highlighting where each succeeds and where it breaks down. Our main message for JVSTA readers is that publishing review knowledge in a symbolic, machine-actionable form alongside PDFs enables reliable expert querying today and provides a foundation for more trustworthy neurosymbolic assistants in future ALD/ALE workflows.



Figure 1: Modeling a survey table as machine-actionable data in the Open Research Knowledge Graph (ORKG). Panel (1) shows Table 5 from the rare-earth ALD review by Ghazy et al. [25]. Panel (2) shows the corresponding ORKG comparison <https://orkg.org/comparisons/R1469991>, where rows of the original table are represented as contribution columns and the original table column headers as ORKG properties. For readability, only three of the nine papers included in the full comparison are displayed. The following seven are modeled properties as web resources: host material matrix, annealing temperature in °C, external quantum efficiency in %, power efficiency $\times 10^{-4}$, threshold voltage in volts, emission lifetime in milliseconds, and operational device lifetime in hours.

2 Background: The Open Research Knowledge Graph

Large industrial knowledge graphs—such as those developed at Google, Microsoft, Facebook, eBay, and IBM—show how structuring information as *entities* and *relations* can transform access to